

Azmarri Sultana

Id 22201949

CSE427

Section 06

TOPIC NAME : Assignment

DAY : _____

TIME : _____

DATE : / /

① a)

SN	Student category	Department	Result
1	Junion	CSE	Average
2	Senion	EEE	Good
3	Senion	CSE	Bad
4	Junion	EEE	Average
5	Senion	EEE	Good
6	Junion	EEE	Bad
7	Junion	CSE	Good
8	Senion	CSE	Good
9	Senion	CSE	Good
10	Junion	EEE	Bad

Student Category

Junion

Senion

①^A ④^A ⑥^B
⑦^G ⑩^B

②^G ③^B ⑤^G
⑧^G ⑨^G

$$\begin{aligned} \text{Gini (Junion)} &= 1 - \left(\frac{2}{9}\right)^2 - \left(\frac{2}{5}\right)^2 - \left(\frac{1}{5}\right)^2 \\ &= 0.64 \end{aligned}$$

GOOD LUCK

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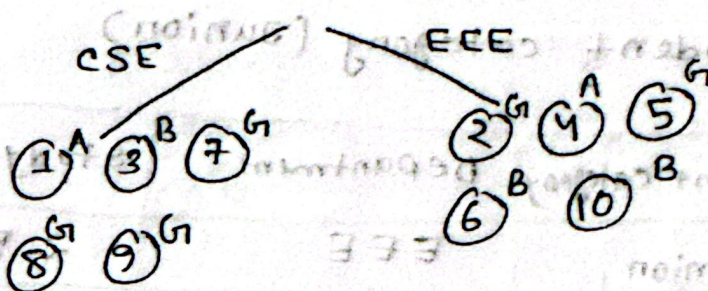
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$$\text{Gini (senior)} = 1 - \left(\frac{0}{5}\right)^2 - \left(\frac{1}{5}\right)^2 - \left(\frac{4}{5}\right)^2$$

$$= 0.32$$

$$\text{Gini (student category)} = \frac{5}{10} \times 0.64 + \frac{5}{10} \times 0.32 = 0.48$$

Department



$$\text{Gini (CSE)} = 1 - \left(\frac{1}{5}\right)^2 - \left(\frac{1}{5}\right)^2 - \left(\frac{3}{5}\right)^2 = 0.56$$

$$\text{Gini (EEE)} = 1 - \left(\frac{1}{5}\right)^2 - \left(\frac{2}{5}\right)^2 - \left(\frac{2}{5}\right)^2 = 0.64$$

$$\text{Gini (department)} = \frac{5}{10} \times 0.56 + \frac{5}{10} \times 0.64 = 0.6$$

$$\text{Gini (student category)} < \text{Gini (department)}$$

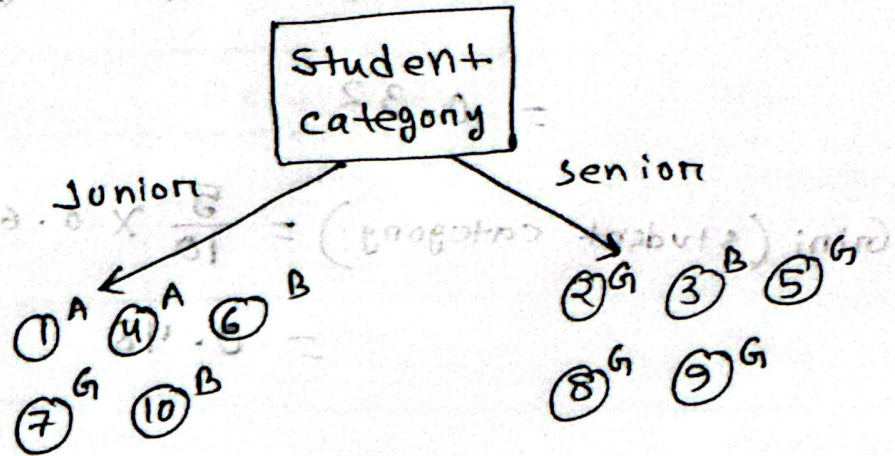
So, the root node is student category. It would be better candidate.

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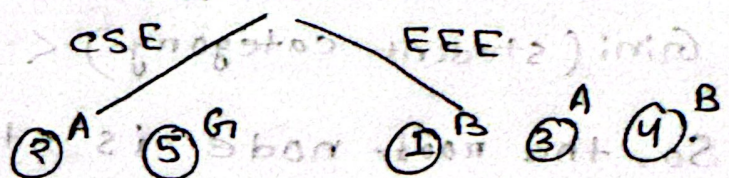
b)



Student category (Junior)

Student category	Department	Study hours	Result
① Junion	EEE	2.5	Bad
② Junion	CSE	3	Average
③ Junion	EEE	4	Average
④ Junion	EEE	4	Bad
⑤ Junion	CSE	5	Good

Department



$$\text{Gini (CSE)} = 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = 0.5$$

$$\text{Gini (EEE)} = 1 - \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2 = 0.44$$

GOOD LUCK

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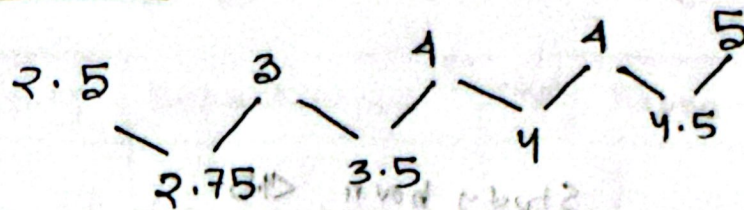
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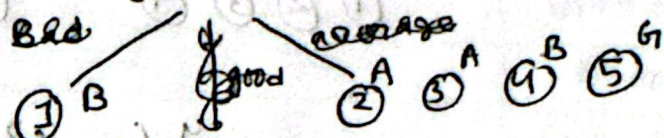
$$\text{Gini (department)} = \frac{2}{5} \times 0.5 + \frac{3}{5} \times 0.44$$

$$= 0.464$$

Study hours

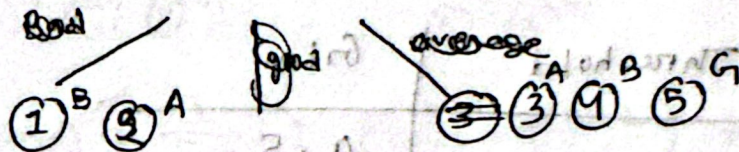


Study hours < 2.75



$$\text{Gini (SH < 2.75)} = \frac{1}{5} \left(1 - \left(\frac{1}{1} \right)^2 \right) + \frac{4}{5} \left(1 - \left(\frac{2}{4} \right)^2 - \left(\frac{1}{4} \right)^2 - \left(\frac{1}{4} \right)^2 \right) = 0.5$$

Study hours < 3.5

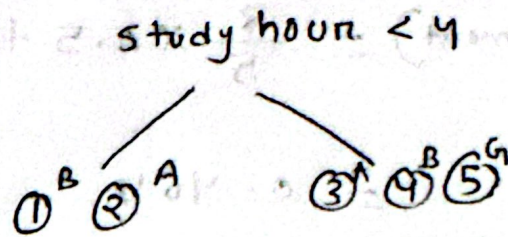


$$\text{Gini (SH < 3.5)} = \frac{2}{5} \left(1 - \left(\frac{1}{2} \right)^2 - \left(\frac{1}{2} \right)^2 \right) + \frac{3}{5} \left(1 - \left(\frac{1}{3} \right)^2 - \left(\frac{1}{3} \right)^2 - \left(\frac{1}{3} \right)^2 \right) = 0.64 + 0.15 = 0.35$$

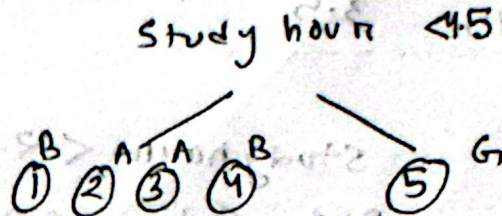
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$$\text{Gini} (SH < 4) = 0.6$$



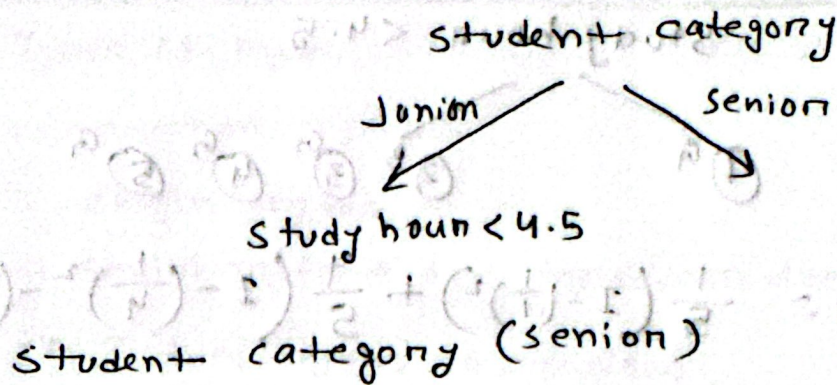
$$\begin{aligned} \text{Gini} (SH < 4.5) &= \frac{4}{5} \left(1 - \left(\frac{2}{4} \right)^2 - \left(\frac{2}{4} \right)^2 \right) + \frac{1}{5} \left(1 - \left(\frac{1}{1} \right)^2 \right) \\ &= 0.4 \end{aligned}$$

Threshold	Gini
2.75	0.5
3.5	0.6
4	0.6
4.5	0.4

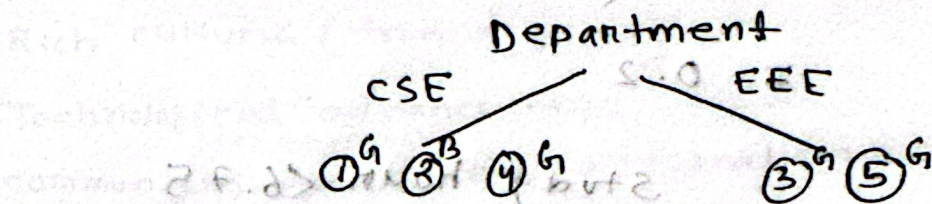
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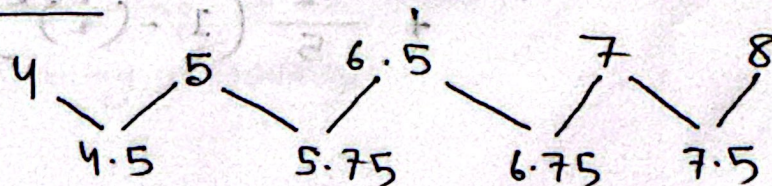
Student Category	Department	Study hours	Result
Senior	CSE	4	Good
Senior	CSE	5	Bad
Senior	EEE	6.5	Good
Senior	CSE	7	Good
Senior	EEE	8	Good



$$\text{Gini (Department)} = \frac{3}{5} \left(1 - \left(\frac{2}{3} \right)^2 - \left(\frac{1}{3} \right)^2 \right) + \frac{2}{5} \left(1 - \left(\frac{2}{2} \right)^2 \right)$$

$$= 0.267$$

Study hours



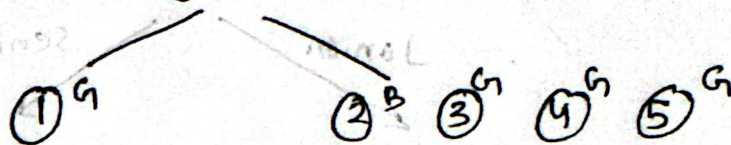
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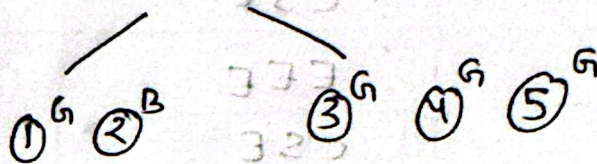
Study hours < 4.5



$$Gini(SH < 4.5) = \frac{1}{5} \left(1 - \left(\frac{1}{5} \right)^2 \right) + \frac{4}{5} \left(1 - \left(\frac{1}{4} \right)^2 - \left(\frac{3}{4} \right)^2 \right)$$

$$= 0.3$$

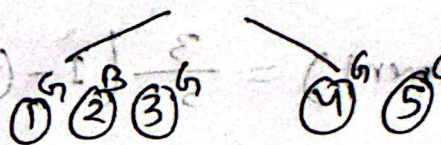
Study hours < 5.75



$$Gini(SH < 5.75) = \frac{2}{5} \left(1 - \left(\frac{1}{2} \right)^2 - \left(\frac{1}{2} \right)^2 \right) + \frac{3}{5} \left(1 - \left(\frac{3}{3} \right)^2 \right)$$

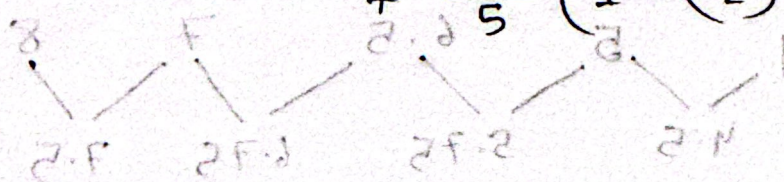
$$= 0.2$$

Study hours < 6.75



$$Gini(SH < 6.75) = \frac{3}{5} \left(1 - \left(\frac{2}{3} \right)^2 - \left(\frac{1}{3} \right)^2 \right)$$

$$+ \frac{2}{5} \left(1 - \left(\frac{2}{2} \right)^2 \right) = 0.267$$

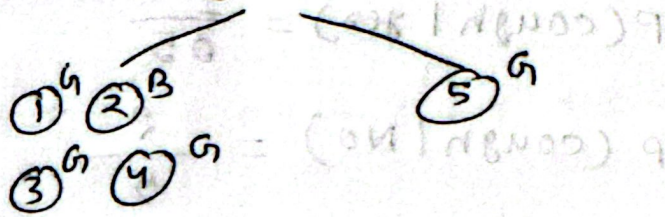


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study hour < 7.5



$$\text{Gini}(\text{SH} < 7.5) = \frac{4}{5} \left(1 - \left(\frac{1}{4} \right)^2 - \left(\frac{3}{4} \right)^2 \right) + \frac{1}{5} \left(1 - \left(\frac{1}{1} \right)^2 \right)$$

$$= 0.3$$

Threshold	Gini
4.5	0.3
5.75	0.2
6.75	0.267
7.5	0.3

Student category

Junior

Senior

Study hour < 4.5

Study hour < 5.75

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③

$$P(\text{cough} | \text{yes}) = \frac{1}{5}$$

$$P(\text{cough} | \text{No}) = \frac{2}{5}$$

$$P(\text{antibiotic} | \text{yes}) = \frac{1}{5}$$

$$P(\text{antibiotic} | \text{No}) = \frac{3}{5}$$

$$P(\text{Female} | \text{yes}) = \frac{4}{5}$$

$$P(\text{Female} | \text{No}) = \frac{1}{5}$$

$$\mu_{\text{yes}} = \frac{101 + 99 + 98 + 97 + 99}{5} = 98.8$$

$$\mu_{\text{No}} = \frac{103 + 102 + 100 + 104 + 101}{5} = 102$$

Temperature (x)	(x - μ)	(x - μ) ²	Recover
101	2.2	4.84	yes
99	0.2	0.04	yes
103	1	1	no
98	-0.8	0.64	yes
102	0	0	no
100	2	4	no
97	-1.8	3.24	yes
104	2	4	no
99	0.2	0.04	yes
101	-1	1	no

GOOD LUCK

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$$\sigma_{yes}^2 = \frac{1}{5} \sum (x - \mu_{yes})^2$$

$$= \frac{1}{5} (4.84 + 0.04 + 0.64 + 3.24 + 0.04)$$

$$= 1.76$$

$$\therefore \sigma_{yes} = 1.32$$

$$\sigma_{No}^2 = \frac{1}{5} \sum (x - \mu_{No})^2$$

$$= \frac{1}{5} (1 + 0 + 4 + 4 + 1) = 2$$

$$\therefore \sigma_{No} = 1.41$$

$$P(T=99 | yes) = \frac{1}{\sigma_{yes} \sqrt{2\pi}} e^{-\left[\frac{(99 - \mu_{yes})^2}{2\sigma_{yes}^2} \right]}$$

$$= \frac{1}{1.32 \sqrt{2\pi}} e^{-\frac{(99 - 98.8)^2}{2 \times 1.32^2}}$$

$$= 0.299$$

$$P(T=99 | No) = \frac{1}{\sigma_{No} \sqrt{2\pi}} e^{-\left[\frac{(99 - \mu_{No})^2}{2\sigma_{No}^2} \right]}$$

$$= \frac{1}{1.41 \sqrt{2\pi}} e^{-\left[\frac{(99 - 102)^2}{2 \times 1.41^2} \right]}$$

$$= 0.0394$$

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$$P(\text{yes} | \text{99}^\circ \text{1 cough 1 Antibiotic 1 Female})$$

$$= P(\text{99}^\circ \text{1 cough 1 Antibiotic 1 Female} | \text{yes}) P(\text{yes})$$

$$= P(\text{99}^\circ | \text{yes}) P(\text{cough} | \text{yes}) P(\text{Antibiotic} | \text{yes})$$

$$P(\text{Female} | \text{yes}) P(\text{yes})$$

$$= 0.299 \times \frac{1}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{5}{10}$$

$$= 4.784 \times 10^{-3}$$

$$P(\text{No} | \text{99}^\circ \text{1 cough 1 antibiotic 1 Female})$$

$$= P(\text{99}^\circ \text{1 cough 1 antibiotic 1 Female} | \text{No}) P(\text{No})$$

$$= P(\text{99}^\circ | \text{No}) P(\text{cough} | \text{No}) P(\text{antibiotic} | \text{No})$$

$$P(\text{Female} | \text{No}) P(\text{No})$$

$$= 0.0294 \times \frac{2}{5} \times \frac{3}{5} \times \frac{1}{5} \times \frac{5}{10}$$

$$= 7.056 \times 10^{-4}$$

The probability of recovery is higher.

GOOD LUCK

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④

a)

FLU = Yes

The sample with 'yes' target has 2 'yes' in the "Cough" feature and 0 "No".

So, Cough = Yes

Given Oxygen Saturation Percentage 98, 94, 92, 96

If we sort it: 92, 94, 96, 98

Average
Median of Oxygen Saturation Percentage:

$$= \frac{92 + 94 + 96 + 98}{4} = 95$$

Sorted age: 22, 28, 30, 45

Average
Median of age = $\frac{22 + 28 + 30 + 45}{4} = 31.25$

So, initial guess for oxygen saturation percentage

$$= 95$$

initial guess for age = 31.25

GOOD LUCK

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b) For Cough,

Weighted frequency of yes = Frequency of yes $\times$ weight of yes

$$= \frac{3}{4} \times \frac{0.35 + 0.40 + 0.10}{0.15 + 0.35 + 0.40 + 0.10}$$

$$= 0.6375$$

Weighted frequency of No = frequency of No $\times$ weight of No

$$= \frac{1}{4} \times \frac{0.15}{0.15 + 0.35 + 0.40 + 0.10}$$

$$\left(\left(\frac{1}{N} \right) - \left(\frac{E}{N} \right) - 1 \right) \frac{1}{10} = 0.0375$$

$$\left(\left(\frac{1}{E} \right) - \left(\frac{E}{N} \right) - 1 \right) \frac{E}{10} = 0.0375$$

$$\therefore \text{Estimated value (cough)} = \text{yes} = 0.6375$$

$$\text{Sum of proximity} = 0.15 + 0.35 + 0.40 + 0.10 = 1$$

Oxygen saturation percentage

$$\text{Weighted average} = (98 \times 0.15) + (94 \times 0.35) + (92 \times 0.40)$$

$$\left(\left(\frac{E}{N} \right) - \left(\frac{E}{N} \right) - 1 \right) \frac{E}{10} = 0.0375$$

$$\text{Weighted average (age)} = (22 \times 0.15) + (30 \times 0.35) + (40 \times 0.40) + (28 \times 0.10) = 34.6$$

Estimated missing values of cough, oxygen, age is yes, 94, 34.6

(5) a)

$$\text{Initial weight} = \frac{1}{N} = \frac{1}{10}$$

(b)

Device colon:

Red (4): Positive = 3, negative = 1

Blue (3): Positive = 0, negative = 3

Green (3): Positive = 2, negative = 1

$$\text{Gini(device colon)} = \frac{4}{10} \left(1 - \left(\frac{3}{4} \right)^2 - \left(\frac{1}{4} \right)^2 \right)$$

$$+ \frac{3}{10} \left(1 - 0 - \left(\frac{3}{3} \right)^2 \right) + \frac{3}{10} \left(1 - \left(\frac{2}{3} \right)^2 - \left(\frac{1}{3} \right)^2 \right)$$

$$= \frac{3}{20} + 0 + \frac{2}{15} = \frac{17}{60} = 0.283$$

Device type:

Mobile (5): Positive (3), negative (2)

Desktop (5): Positive = 2, negative = 3

$$\text{Gini(device type)} = \frac{5}{10} \left(1 - \left(\frac{3}{5} \right)^2 - \left(\frac{2}{5} \right)^2 \right)$$

$$+ \frac{5}{10} \left(1 - \left(\frac{2}{5} \right)^2 - \left(\frac{3}{5} \right)^2 \right)$$

$$= 0.48$$

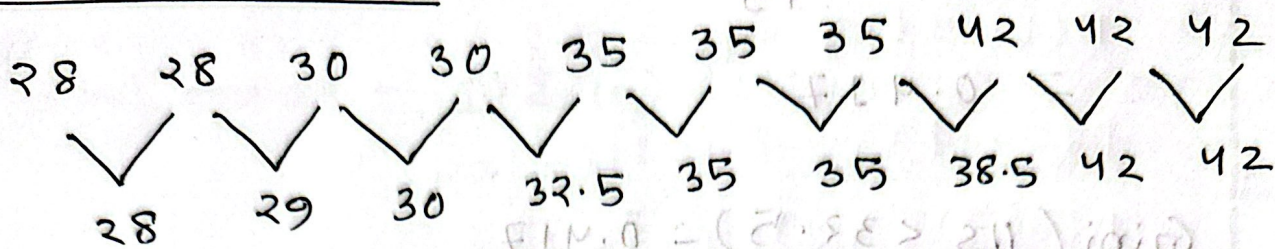
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Usability Score:



Usability score ≤ 28

$$- (1) + (2) + (1) \quad 4 - 5 \quad 1 + 2 - 3 + 6 + 7 + 8 - 9 - 10 +$$

$$Gini(US \leq 28) = \frac{2}{10} \left(1 - \left(\frac{2}{2} \right)^2 \right) + \frac{8}{10} \left(1 - \left(\frac{5}{8} \right)^2 - \left(\frac{3}{8} \right)^2 \right)$$

$$= 0.375$$

Usability score < 29

$$4 - 5 \quad 1 + 2 - 3 + 6 + 7 + 8 - 9 - 10 +$$

$$Gini(US < 29) = 0.375$$

Usability score ≤ 30

$$4 - 5 \quad 1 + 2 - 3 + 6 + 7 + 8 - 9 - 10 +$$

$$Gini(US \leq 30) = \frac{4}{10} \left(1 - \left(\frac{1}{4} \right)^2 - \left(\frac{3}{4} \right)^2 \right) + \frac{6}{10} \left(1 - \left(\frac{4}{6} \right)^2 - \left(\frac{2}{6} \right)^2 \right)$$

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$$= \frac{3}{20} + \frac{4}{15}$$

$$= 0.417$$

$$\text{Gini} (US < 32.5) = 0.417$$

Usability Score ≤ 35

$$\begin{array}{l} \textcircled{2} - \textcircled{3} + \textcircled{4} - \textcircled{5} - \textcircled{1} + \textcircled{6} + \textcircled{9} - \\ \textcircled{7} + \textcircled{8} - \textcircled{10} + \end{array}$$

$$\text{Gini} (US \leq 35) = \frac{7}{10} \left(1 - \left(\frac{3}{7} \right)^2 - \left(\frac{4}{7} \right)^2 \right) + \frac{3}{10} \left(1 - \left(\frac{2}{3} \right)^2 - \left(\frac{1}{3} \right)^2 \right)$$

$$= \frac{12}{35} + \frac{2}{15}$$

$$= 0.476$$

$$\text{Gini} (US < 38.5) = 0.476$$

$$\text{Gini} (US \leq 42) = \frac{10}{10} \left(1 - \left(\frac{5}{10} \right)^2 - \left(\frac{5}{10} \right)^2 \right)$$

$$= 0.5$$

lowest Gini is device colon.

So the first stump is device colon.

GOOD LUCK

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c)

$$\begin{aligned} \text{Total error} &= 1(1/10) + 0(1/10) + 1(1/10) \\ &= \frac{1}{5} = 0.2 \end{aligned}$$

$$\begin{aligned} \text{amount of sag } (\alpha) &= \frac{1}{2} \ln \left(\frac{1 - \text{total error}}{\text{total error}} \right) \\ &= 0.69 \end{aligned}$$

d) amount of sag $(\alpha) = 0.69$

incorrectly classified:

$$\begin{aligned} \text{updated weight} &= \text{sample weight} \times e^{\alpha} \\ &= \frac{1}{10} \times e^{0.69} \\ &= 0.199 \end{aligned}$$

correctly classified:

$$\begin{aligned} \text{updated weight} &= \text{sample weight} \times e^{-\alpha} \\ &= \frac{1}{10} \times e^{-0.69} \\ &= 0.0502 \end{aligned}$$

⑥

$$P(y=1|x) = \sigma(z) = \hat{y}$$

$$P(y=0|x) = 1 - \sigma(z) = 1 - \hat{y}$$

$$P(y|x) = (\hat{y})^y (1-\hat{y})^{1-y} =$$

If $y=1$,

$$P(y=1|x) = (\hat{y})^1 \times (1-\hat{y})^{1-1} = \hat{y}^1$$

If $y=0$,

$$P(y=0|x) = (\hat{y})^0 \times (1-\hat{y})^{1-0} = (1-\hat{y})$$

Total probability / Likelihood:

$$P(y^1, y^2, y^3, \dots, y^m | x^1, x^2, x^3, \dots, x^m)$$

$$= P(y^1|x^1) P(y^2|x^2) P(y^3|x^3) \dots P(y^m|x^m)$$

$$= \prod_{i=1}^m P(y^{(i)} | x^{(i)})$$

$$= \prod_{i=1}^m (\hat{y}^{(i)})^{y^{(i)}} \times (1-\hat{y}^{(i)})^{1-y^{(i)}}$$

$$\log(\text{likelihood}) = \sum_{i=1}^m \log [(\hat{y}^{(i)})^{y^{(i)}} \times (1-\hat{y}^{(i)})^{1-y^{(i)}}]$$

$$= \sum_{i=1}^m \log(\hat{y}^{(i)})^{y^{(i)}} + \log(1-\hat{y}^{(i)})^{1-y^{(i)}}$$

$$= \sum_{i=1}^m y^{(i)} \log(\hat{y}^{(i)}) + (1-y^{(i)}) \log(1-\hat{y}^{(i)})$$

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We need to minimize cross entropy loss function,

$$\text{Loss function (LCE)} = - \sum_{i=1}^m [y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})]$$

For updating weight, we need derivative of loss function = $\frac{\partial \text{LCE}}{\partial w_i}$

$$z = w_i x_i + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$w_i = w_i - \eta \frac{\partial \text{LCE}}{\partial w_i}$$

$$\frac{\partial \text{LCE}}{\partial w_i} = \frac{\partial \text{LCE}}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial w_i}$$

$$\frac{\partial \text{LCE}}{\partial \hat{y}} = - \frac{\partial}{\partial \hat{y}} (y \log \hat{y}) - \frac{\partial}{\partial \hat{y}} [(1 - y) \log(1 - \hat{y})]$$

$$= - \frac{y}{\hat{y}} - (1 - y) \times \frac{1}{1 - \hat{y}} \times \frac{\partial}{\partial \hat{y}} (1 - \hat{y})$$

$$= - \frac{y}{\hat{y}} - \frac{1 - y}{1 - \hat{y}} (-1) = \frac{-y + y\hat{y} + 1 - y\hat{y}}{\hat{y}(1 - \hat{y})} = \frac{\hat{y} - y}{\hat{y}(1 - \hat{y})}$$

$$\frac{\partial \hat{y}}{\partial z} = \frac{\partial \sigma(z)}{\partial z} = \sigma(z) (1 - \sigma(z)) = \hat{y} (1 - \hat{y})$$

$$\frac{\partial z}{\partial w_i} = \frac{\partial}{\partial w_i} (w_i x_i + b) = x_i$$

$$\frac{\partial \text{LCE}}{\partial w_i} = \frac{\hat{y} - y}{\hat{y}(1 - \hat{y})} \times \hat{y}(1 - \hat{y}) x_i = (\hat{y} - y) x_i$$

$$\therefore w_i(\text{new}) = w_i(\text{old}) - \eta (\hat{y} - y) x_i$$